SQL BUSINESS CASE STUDY- TARGET BRAZIL

● Target is a globally renowned brand and a prominent retailer in the United States. Target makes itself a preferred shopping destination by offering outstanding value, inspiration, innovation and an exceptional guest experience that no other retailer can deliver.

● This particular business case focuses on the operations of Target in Brazil and provides insightful information about 100,000 orders placed between 2016 and 2018. The dataset offers a comprehensive view of various dimensions including the order status, price, payment and freight performance, customer location, product attributes, and customer reviews.

● By analyzing this extensive dataset, it becomes possible to gain valuable insights into Target's operations in Brazil. The information can shed light on various aspects of the business, such as order processing, pricing strategies, payment and shipping efficiency, customer demographics, product characteristics, and customer satisfaction levels.

1.EXPLORATORY ANALYSIS

1.a Information Schema

```
SELECT column_name, data_type
FROM `scalersqlpractice-460317.sql_business_case.INFORMATION_SCHEMA.COLUMNS`
WHERE table_name = 'customers';
```

Row	column_name	data_type
1	customer_id	STRING
2	customer_unique_id	STRING
3	customer_zip_code_prefix	INT64
4	customer_city	STRING
5	customer_state	STRING

Insights

There are 5 main columns in the customers table where customer_id , customer_unique_id, customer_city, and customer_state are of string datatype. There is one customer_zip_code_prefix which defines the zipcode of customer city and it is an integer datatype.

1.b Time range of orders

```
  max(order_purchase_timestamp) as last_order,
  min(order_purchase_timestamp) as first_order
from `scalersqlpractice-460317.sql_business_case.orders`;
```

Row	last_order	first_order
1	2018-10-17 17:30:18 UTC	2016-09-04 21:15:19 UTC

Insights

The first order was given on 4th September 2016 at 21:15 pm and the last order was given on 17th October 2018 at 17:30 pm . The dataset covers the data along these years from 2016 to 2018.

1.c Count of unique cities and states

```
select count(distinct customer_city) as unique_city,
       count(distinct customer_state) as unique_state
from `scalersqlpractice-460317.sql_business_case.customers`;
```

Row	unique_city	unique_state
1	4119	27

Insights

There are 4119 unique cities and 27 states where Target Brazil has its customers.

2. Trends and Seasonality:

2.a Growing trend of orders (yearly/monthly)

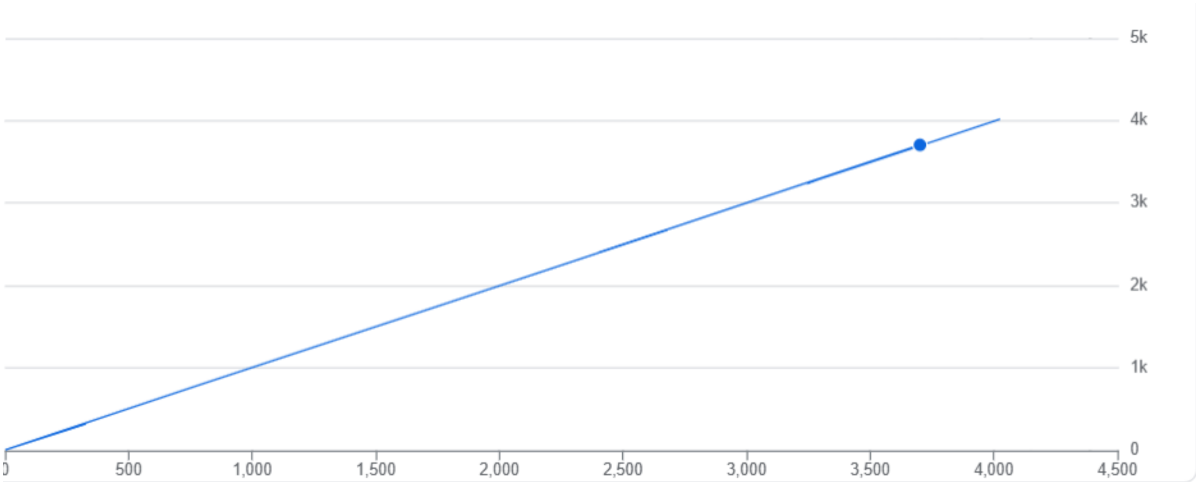
```
select
    EXTRACT(YEAR FROM order_purchase_timestamp) AS year,
    EXTRACT(MONTH FROM order_purchase_timestamp) AS month,
    COUNT(order_id) AS total_orders

from `scalersqlpractice-460317.sql_business_case.orders`
group by 1,2
order by year asc,month asc
limit 10;
```

Row	year	month	total_orders
1	2016	9	4
2	2016	10	324
3	2016	12	1
4	2017	1	800
5	2017	2	1780
6	2017	3	2682
7	2017	4	2404
8	2017	5	3700
9	2017	6	3245
10	2017	7	4026

Insights

There is a steady increase in total number of order since 2016 to 2018



2.b Monthly Seasonality

select

extract(month from order_purchase_timestamp) as months, count(order_id) as total_orders

from `scalersqlpractice-460317.sql_business_case.orders`

where order_status = 'delivered'

group by 1

order by total_orders desc;

Row	months	total_orders
1	8	10544
2	5	10295
3	7	10031
4	3	9549
5	6	9234
6	4	9101
7	2	8208
8	1	7819
9	11	7289
10	12	5514
11	10	4743

Insights

The months of August, May , and July are the highest months where the customers have orders the most number of times. Meanwhile, September, October, and December were the months where we have the least number of Orders.

2.c Orders by Time and Day

SELECT

CASE

WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 0 AND 6 THEN 'Dawn'

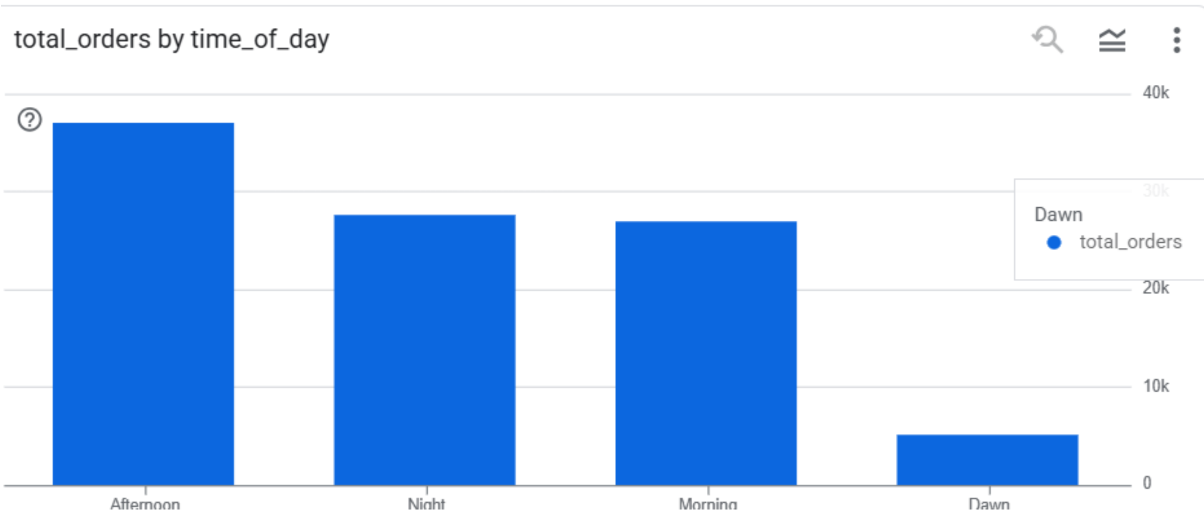
WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 7 AND 12 THEN 'Morning'

WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 13 AND 18 THEN 'Afternoon'

ELSE 'Night'

```
 END AS time_of_day,
 COUNT(order_id) AS total_orders
FROM `scalersqlpractice-460317.sql_business_case.orders`
WHERE order_status = 'delivered'
GROUP BY time_of_day
ORDER BY total_orders DESC;
```

Row	time_of_day	total_orders
1	Afternoon	36965
2	Night	27522
3	Morning	26919
4	Dawn	5072



Insights

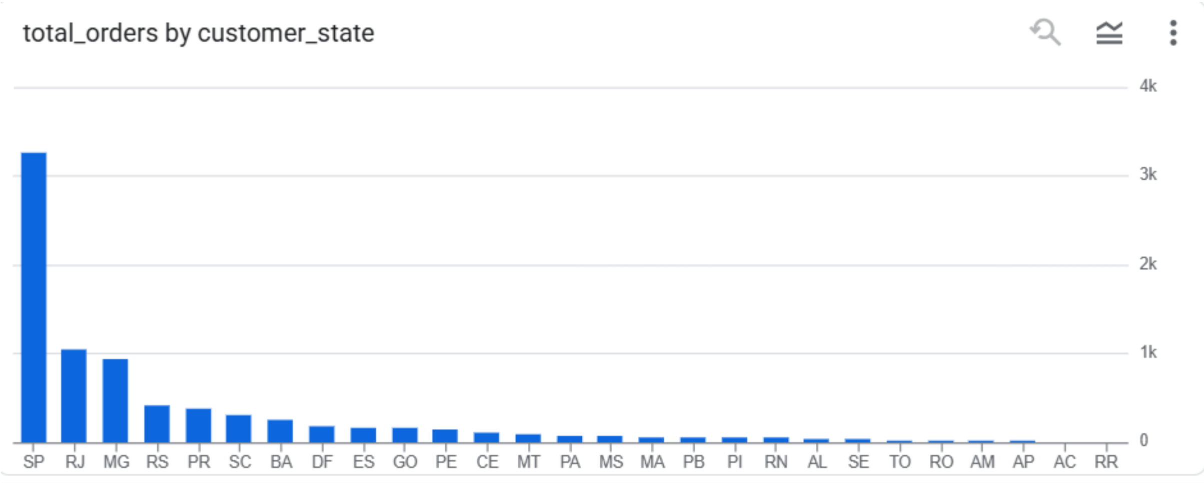
According to the results the Afternoon period and Night time period are the major chunks of orders were ordered by the customers where Afternoon period stands significantly higher than any other time period i.e. 9,443 more orders than Night period.

3. Evolution of E-commerce orders in the Brazil region:

3.a Month-on-month orders by state

```
select
    customer_state,
    extract(year from order_purchase_timestamp) as year,
    extract(month from order_purchase_timestamp) as month,
    count(o.order_id) as total_orders
from `scalersqlpractice-460317.sql_business_case.orders` o join
  `scalersqlpractice-460317.sql_business_case.customers` c
  on o.customer_id = c.customer_id
group by 1,2,3
order by total_orders desc
limit 10;
```

Row	customer_state	year	month	total_orders
1	SP	2018	8	3253
2	SP	2018	5	3207
3	SP	2018	4	3059
4	SP	2018	1	3052
5	SP	2018	3	3037
6	SP	2017	11	3012
7	SP	2018	7	2777
8	SP	2018	6	2773
9	SP	2018	2	2703
10	SP	2017	12	2357



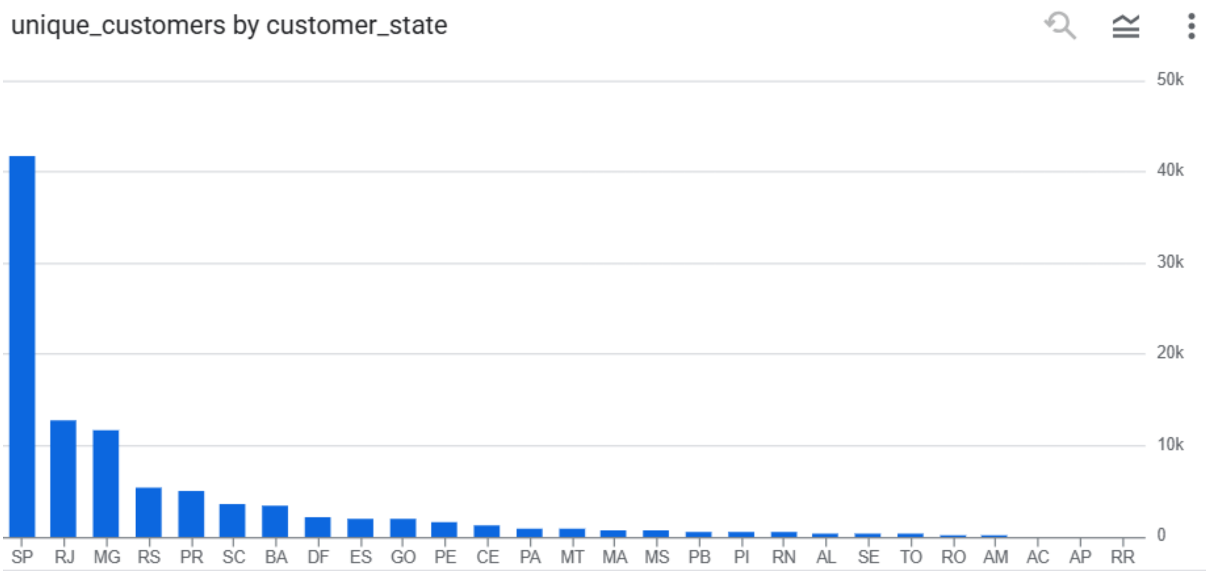
Insights

According to the following results we see that Sao Paolo, Rio De Janeiro, and Minas Gerais are the top 3 states where most number of orders given by customers.

3.b Customer distribution across all the states

```
SELECT
  c.customer_state,
  COUNT(DISTINCT c.customer_id) AS unique_customers
FROM `scalersqlpractice-460317.sql_business_case.customers` c
GROUP BY c.customer_state
ORDER BY unique_customers DESC;
```

Row	customer_state	unique_customers
1	SP	41746
2	RJ	12852
3	MG	11635
4	RS	5466
5	PR	5045
6	SC	3637
7	BA	3380
8	DF	2140
9	ES	2033
10	GO	2020



Insights

The results shows us that majority of customers for target are from top 3 major states i.e. Sao Paolo, Rio de Janeiro, and Minas Gerais.

4. Impact on Economy:

4.a % increase in order value (2017 → 2018, Jan–Aug only)

```
with yearly_sales as(
  select
  extract(year from order_purchase_timestamp) as year,
  SUM(p.payment_value) AS total_sales
  from `scalersqlpractice-460317.sql_business_case.orders` o
  join `scalersqlpractice-460317.sql_business_case.payments` p
  on o.order_id =p.order_id
 WHERE EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8
    AND o.order_status = 'delivered'
  GROUP BY year

)

SELECT
  year,
  total_sales,
  LAG(total_sales) OVER (ORDER BY year) AS prev_year_sales,
  ROUND(((total_sales - LAG(total_sales) OVER (ORDER BY year)) / LAG(total_sales) OVER (ORDER BY year)) * 100, 2) AS pct_increase
FROM yearly_sales;
```

Row	year	total_sales	prev_year_sales	pct_increase
1	2017	3473862.759999...	null	null
2	2018	8452975.200000...	3473862.759999...	143.33

Insights

Massive Growth YoY (2017 → 2018)

- Sales increased from **3.47M in 2017** to **8.45M in 2018**.
- That's a **143% jump**, meaning sales **more than doubled** in just one year.

Customer/Order Growth Likely

- Such a sharp increase is usually driven by either:

- A big increase in **number of customers/orders**, or
- Higher **average order value (AOV)**, possibly due to bigger-ticket items or more cross-sell/upsell.

Seasonality Control

- Since you restricted data to **Jan–Aug (months 1–8)**, this is a **like-for-like period comparison**, which makes the growth even more reliable.

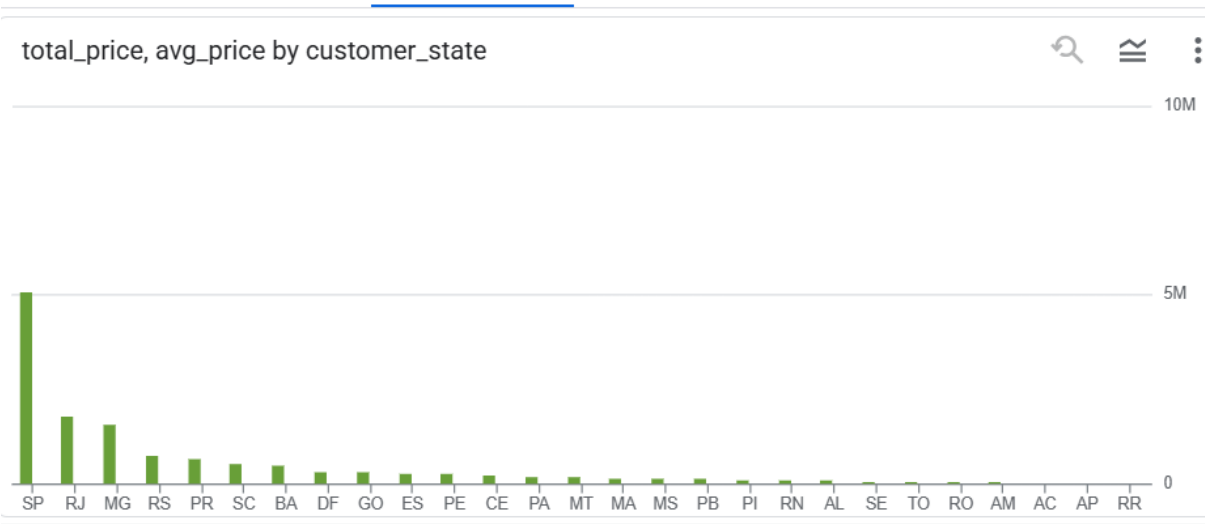
Healthy Business Momentum

- A 143% YoY increase indicates **strong demand and scaling ability**.

4.b Total & average order price per state

```
SELECT
  c.customer_state,
  SUM(oi.price) AS total_price,
  AVG(oi.price) AS avg_price
FROM `scalersqlpractice-460317.sql_business_case.orders` o
JOIN `scalersqlpractice-460317.sql_business_case.customers` c ON o.customer_id = c.customer_id
JOIN `scalersqlpractice-460317.sql_business_case.order_items` oi ON o.order_id = oi.order_id
WHERE o.order_status = 'delivered'
GROUP BY c.customer_state
ORDER BY total_price DESC;
```

Row	customer_state	total_price	avg_price
1	SP	5067633.160001...	109.1033663451...
2	RJ	1759651.129999...	124.4185201159...
3	MG	1552481.829999...	120.1983454629...
4	RS	728897.470000021	118.8290626018...
5	PR	666063.5100000...	117.9082156133...



Insights

SP is the revenue leader

- São Paulo contributes **~5.07M**, which is almost **3x more** than the next state (RJ).
- This suggests that SP is the **core revenue driver** and likely the largest customer base.

High-value orders in RJ & MG

- RJ and MG have **higher avg order values** (124 and 120) compared to SP (109).
- Customers here may be purchasing **higher-ticket items** or **larger baskets per order**.

RS & PR are smaller contributors

- RS (728K) and PR (666K) together contribute less than 15% of SP's sales.
- Still, their avg order values (~118) are **healthy**, meaning fewer customers but **quality purchases**.

SP = Volume Game, RJ/MG = Value Game

- SP dominates by sheer **volume of orders**, not necessarily higher spend per order.
- RJ & MG, while smaller in total sales, show **premium customer behavior**.

4.c Total & average freight per state

```
SELECT
 c.customer_state,
 SUM(oi.freight_value) AS total_freight,
 AVG(oi.freight_value) AS avg_freight
```

```
FROM `scalersqlpractice-460317.sql_business_case.orders` o
JOIN `scalersqlpractice-460317.sql_business_case.customers` c ON o.customer_id = c.customer_id
JOIN `scalersqlpractice-460317.sql_business_case.order_items` oi ON o.order_id = oi.order_id
WHERE o.order_status = 'delivered'
GROUP BY c.customer_state
ORDER BY total_freight DESC;
```

Row	customer_state	total_freight	avg_freight
1	SP	702069.9900000...	15.11518235446...
2	RJ	295750.4400000...	20.91143604610...
3	MG	266409.8400000...	20.62634252090...
4	RS	132575.3199999...	21.61319204434...
5	PR	115645.2899999...	20.47181625066...

Insights

□ SP has the highest total freight cost but lowest avg. freight

- **Total Freight:** 702K (almost 2.5x higher than RJ).
- **Avg Freight/order:** Only **15.12**, much lower than all other states.
- This confirms SP has **very high order volumes** but benefits from **logistics efficiency** (shorter distances, better infrastructure, or larger order consolidation).

□ RJ, MG, RS, PR → Higher per-order freight costs

- RJ (~20.91), MG (~20.63), RS (~21.61), PR (~20.47) all have **similar per-order freight**, ~35% higher than SP.
- This suggests deliveries outside SP are **more costly per order**, possibly due to longer distances or weaker logistics networks.

□ RS has the highest avg. freight (21.61)

- This may indicate **longer delivery distances** or **fewer logistics hubs**, driving per-order costs up.

□ RJ vs MG comparison

- RJ has slightly higher **total freight** than MG, despite similar avg freight.
- Suggests RJ processes more orders than MG but at similar cost efficiency.

5. Delivery & Logistics:

5.a Delivery time & estimated vs actual

```
SELECT
  order_id,
  DATE_DIFF(order_delivered_customer_date, order_purchase_timestamp, DAY) AS time_to_deliver,
  DATE_DIFF(order_estimated_delivery_date, order_delivered_customer_date, DAY) AS
diff_estimated_delivery
FROM `scalersqlpractice-460317.sql_business_case.orders`
WHERE order_status = 'delivered'
LIMIT 10;
```

Row	order_id	time_to_deliver	diff_estimated_d...
1	bfbd0f9bdef84302105ad712db...	54	-36
2	3b697a20d9e427646d9256791...	23	0
3	be5bc2f0da14d8071e2d45451...	24	10
4	cd3b8574c82b42fc8129f6d502...	10	39
5	31b0dd6152d2e471443debf03...	8	40
6	d207cc272675637bfed0062edf...	27	22
7	98974b076b01553d49ee64679...	43	6
8	c4b41c36dd589e901f6879f25a...	36	14
9	a8c3f65a43b2e956cbfca10db4...	15	34
10	79ffdd52a918bbe867895a4b18...	22	28

Insights

- **Delivery Time Ranges**: Actual delivery times vary widely (8 to 54 days).
- **Estimation Gaps**: The difference between estimated and actual is often very large (from -36 to +40).

□ **Systemic Issue**: Since both early and late deliveries exist, it's less about logistics only and more about **estimation accuracy**.

□ **Customer Impact**:

- **Early deliveries** = good surprise, but **overestimated times** may discourage purchases.
- **Late deliveries** = negative customer experience, more serious issue.

5.b Top 5 & bottom 5 states by average freight

TOP 5

```
SELECT
  c.customer_state,
  AVG(oi.freight_value) AS avg_freight
FROM `scalersqlpractice-460317.sql_business_case.orders` o
JOIN `scalersqlpractice-460317.sql_business_case.customers` c ON o.customer_id = c.customer_id
JOIN `scalersqlpractice-460317.sql_business_case.order_items` oi ON o.order_id = oi.order_id
WHERE o.order_status = 'delivered'
GROUP BY c.customer_state
ORDER BY avg_freight DESC
LIMIT 5;
```

Row	customer_state	avg_freight
1	PB	43.09168941979...
2	RR	43.08804347826...
3	RO	41.33054945054...
4	AC	40.04791208791...
5	PI	39.11508604206...

BOTTOM 5

```
SELECT
  c.customer_state,
  AVG(oi.freight_value) AS avg_freight
```

```
FROM `scalersqlpractice-460317.sql_business_case.orders` o
JOIN `scalersqlpractice-460317.sql_business_case.customers` c ON o.customer_id = c.customer_id
JOIN `scalersqlpractice-460317.sql_business_case.order_items` oi ON o.order_id = oi.order_id
WHERE o.order_status = 'delivered'
GROUP BY c.customer_state
ORDER BY avg_freight ASC
LIMIT 5;
```

Row	customer_state	avg_freight
1	SP	15.11518235446...
2	PR	20.47181625066...
3	MG	20.62634252090...
4	RJ	20.91143604610...
5	DF	21.07216135881...

Insights

Logistics Advantage in SP

- SP’s freight efficiency (15.1) is driven by:
  - Large customer concentration.
  - Better infrastructure & logistics hubs.
  - Shorter average delivery distances.

Other Southeastern States (RJ, MG, PR, DF)

- Relatively moderate freight (~20–21).
- Still efficient compared to North/Northeast regions.

Northern & Northeastern States (PB, RR, RO, AC, PI)

- Freight cost is **double to triple** that of SP.
- Likely due to:
  - Long distances from main hubs (SP/SE region).

    - Limited delivery networks.
    - Higher last-mile logistics cost.

 Profitability Risk

- If average order value in these high-freight states is not proportionally higher, margins shrink heavily.
- Example: If avg order value is ~120 but avg freight is ~40 → freight = **33% of order value**, vs **14% in SP**.

5.c Top 5 & bottom 5 states by delivery time

```
SELECT
  c.customer_state,
  AVG(DATE_DIFF(o.order_delivered_customer_date, o.order_purchase_timestamp, DAY)) AS avg_delivery_time
FROM `scalersqlpractice-460317.sql_business_case.orders` o
JOIN `scalersqlpractice-460317.sql_business_case.customers` c ON o.customer_id = c.customer_id
WHERE o.order_status = 'delivered'
GROUP BY c.customer_state
ORDER BY avg_delivery_time DESC
LIMIT 5; -- slowest states
```

Row	customer_state	avg_delivery_time
1	RR	28.97560975609...
2	AP	26.73134328358...
3	AM	25.98620689655...
4	AL	24.04030226700...
5	PA	23.31606765327...

Insights

 Very Long Delivery Times in Northern States

- RR, AP, and AM (all in the **North region**) have the **longest delivery times**, ranging from **26–29 days**.

- This reflects logistical challenges: long distances from distribution centers (mainly in SP/SE region), reliance on air/river transport, and fewer logistics hubs.

? **Northeast States Still Slow**

- AL (24 days) and PA (23 days) also face long delivery times, though slightly better than Northern states.
- Indicates a systemic gap in **North & Northeast delivery infrastructure** compared to Southeast (SP/RJ), which usually delivers in under ~10 days.

? **Customer Experience Risk**

- Delivery times approaching **a month** significantly impact customer satisfaction and repurchase behavior.
- If competitors offer faster delivery, customers in these regions may churn.

5.d Fastest delivery compared to estimated

```
SELECT
  c.customer_state,
  AVG(DATE_DIFF(o.order_estimated_delivery_date, o.order_delivered_customer_date, DAY)) AS avg_days_early
FROM `scalersqlpractice-460317.sql_business_case.orders` o
JOIN `scalersqlpractice-460317.sql_business_case.customers` c ON o.customer_id = c.customer_id
WHERE o.order_status = 'delivered'
GROUP BY c.customer_state
ORDER BY avg_days_early DESC
LIMIT 5;
```

Row	customer_state	avg_days_early
1	AC	19.7625
2	RO	19.13168724279...
3	AP	18.73134328358...
4	AM	18.60689655172...
5	RR	16.41463414634...

Insights

Northern States See Huge Early Deliveries

- Orders in **AC, RO, AP, AM, and RR** are delivered **16–20 days earlier** than promised.
- This suggests the **estimated delivery timeframes for these regions are highly over-padded**.

Overestimation of Delivery Time

- For example, if the system promises **30 days**, actual deliveries may happen in **10–12 days**, creating an "early" appearance of 18–20 days.
- This indicates **poor accuracy in delivery time predictions** rather than exceptional logistics.

Customer Perception

- While customers may be pleasantly surprised by early deliveries, excessively padded delivery windows can **discourage purchases** upfront (if they see "Delivery in 30 days").

Contrast with Actual Delivery Delays

- From your earlier table, states like RR had **high actual delivery times (28.9 days)**, yet they also appear **16 days early** here.
- This contradiction highlights that **promised delivery dates are unrealistic** (system predicts ~45 days+, but actual is ~29).

6. Payments

6.a Month-on-month orders by payment type

```
SELECT
  EXTRACT(YEAR FROM o.order_purchase_timestamp) AS year,
  EXTRACT(MONTH FROM o.order_purchase_timestamp) AS month,
  p.payment_type,
  COUNT(DISTINCT o.order_id) AS total_orders
FROM `scalersqlpractice-460317.sql_business_case.orders` o
JOIN `scalersqlpractice-460317.sql_business_case.payments` p ON o.order_id = p.order_id
WHERE o.order_status = 'delivered'
GROUP BY year, month, p.payment_type
ORDER BY year, month;
```

Row	year	month	payment_type	total_orders
1	2016	10	UPI	51
2	2016	10	credit_card	208
3	2016	10	debit_card	2
4	2016	10	voucher	9
5	2016	12	credit_card	1
6	2017	1	UPI	188
7	2017	1	credit_card	541
8	2017	1	debit_card	9
9	2017	1	voucher	32
10	2017	2	UPI	371
11	2017	2	credit_card	1249

Insights

⍰ **Credit Card Dominates**

- In every month, **credit card is the most preferred payment mode** (208 in Oct '16 → 541 in Jan '17 → 1249 in Feb '17).
- Share of credit card orders is ~70–77%.

⍰ **UPI is Rapidly Growing**

- UPI starts with 51 orders (Oct '16) → jumps to 188 (Jan '17) → 371 (Feb '17).
- UPI adoption is **growing faster than any other payment method**, reflecting the early growth of UPI post-2016 launch.

⍰ **Debit Card & Voucher Negligible**

- Debit Card: Very low usage (2 → 9 orders).
- Voucher: Small but non-zero (9 → 32 orders). Likely used for promotions, not a main method.

⍰ **Order Volume Trend**

- **Total orders are increasing sharply over time**:
    - Oct 2016 → 270 orders
    - Jan 2017 → 770 orders (3× growth)
    - Feb 2017 → 1620 orders (6× growth in 4 months)
- Business is **scaling fast**, with payment preferences consolidating.

6.b Orders by payment Installments

```
SELECT
  p.payment_installments,
  COUNT(DISTINCT o.order_id) AS total_orders
FROM `scalersqlpractice-460317.sql_business_case.orders` o
JOIN `scalersqlpractice-460317.sql_business_case.payments` p ON o.order_id = p.order_id
WHERE o.order_status = 'delivered'
  AND p.payment_installments >= 1
GROUP BY p.payment_installments
ORDER BY p.payment_installments;
```

Row	payment_installm...	total_orders
1	1	47586
2	2	12052
3	3	10147
4	4	6882
5	5	5090
6	6	3800
7	7	1560
8	8	4122
9	9	618
10	10	5137

Insights

Majority Prefer Full Payment

- **47,586 orders (~51%)** are paid in a **single installment**.
- Indicates strong customer preference or ability to pay upfront.

Short-Term Installments (2–3) Are Popular

- 2 installments: 12,052 orders (~13%).
- 3 installments: 10,147 orders (~11%).
- Suggests many customers use installments for **short-term flexibility**.

Longer Installments (6–10) Have Lower Demand

- Usage drops significantly beyond 5 installments.
- Installments 7–9 are **least preferred** (only 1,560–618 orders).

Slight Uptick at 10 Installments

- Interestingly, 10-installment plans (5,137 orders) are more popular than 7–9.
- This suggests customers either want **very short or maximum spread-out payments**, but not mid-range options.

Recommendations for Target Brazil (2016–2018 E-commerce Analysis)

1. Customer & Market Expansion

- Focus on **São Paulo, Rio de Janeiro, and Minas Gerais**, as these states drive the majority of orders.
- Run **localized promotions and targeted digital ads** in underrepresented states (North/Northeast) to expand market share.
- Launch **new user acquisition campaigns** in cities with rising order volumes but smaller customer bases.

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2. Pricing & Payment Strategy

- Introduce **free shipping thresholds** (e.g., free shipping for orders above BRL X) to increase Average Order Value (AOV).
- Offer **0% installment plans** for high-value products, since Brazilians frequently use 3–6 payment installments.
- Provide **discounts on upfront (one-shot) payments** to improve cash flow and reduce dependency on installment payments.
- Educate and incentivize **boleto (bank slip) users** to shift towards digital/credit card payments for higher retention.

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3. Logistics & Delivery

- Establish **regional distribution centers** in the North and Northeast to reduce high freight costs and long delivery delays.
- Strengthen **last-mile delivery partnerships** in remote regions to improve reliability.
- In core regions (SP, RJ), **promote "fast delivery" as a competitive advantage**, since deliveries are already faster than estimates.
- Optimize **delivery time commitments** by aligning estimated dates with actual performance to improve customer trust.

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4. Sales & Seasonal Strategy

- Increase **marketing spend in Q4 (Nov–Dec)** to capture peak demand during Black Friday and holiday shopping.
- Run **Afternoon flash sales** (13:00–18:00), when order volume is highest.
- Explore **seasonal product bundles** (e.g., electronics + accessories) to leverage festive demand spikes.

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5. Operational & Business Efficiency

- Negotiate with **third-party logistics (3PL) partners** to lower freight charges in high-cost states.
- Use **data-driven demand forecasting** to stock warehouses efficiently before seasonal peaks.
- Monitor **regional differences in AOV and freight ratio** to fine-tune pricing and logistics strategies.
- Regularly **analyze delivery performance KPIs** (avg delivery time, freight %, delays) to identify areas needing intervention.